

Co-Optimization of Interdependent Water and Power Distribution Networks

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Abstract—This paper formulates the optimal water and power flow (OWPF) problem to optimize the resources across power and water distribution networks while simultaneously accounting for the coupling between the two infrastructures. The pump schedules in water distribution networks (WDNs) and PV inverters in power distribution networks (PDNs) are jointly managed to minimize the pump power consumption (in WDNs) and active power losses (in PDNs). The resulting OWPF problem includes nonconvex constraints caused by WDN hydraulics and is hard to solve. In order to overcome this problem, we use a novel successive linear approximation-based approach. The method amounts to solving a series of convex optimization problems in WDNs augmented by the LinDistFlow in PDNs—enabling to formulate the problem, which is a mixed-integer quadratically constrained quadratic program called C-OWPF. A case study illustrates the C-OWPF approach's benefits in joint objective improvements with respect to decoupled designs where the PDNs and WDNs are independently operated.

Index Terms—Power distribution networks, water distribution networks, optimization, water-energy nexus, optimal power flow, successive approximation

I. INTRODUCTION

Power and water distribution networks (PDNs and WDNs) are critical infrastructures which are designed and operated independently as separate systems. Nevertheless, these two networks are intrinsically interdependent due to electric-motor-driven pumps in WDNs. Moreover, according to the statistics [1], in the United States, approximately 4% of the total electricity consumption is attributed to the activities of the water network, of which 95%-98% is consumed by water pumps. Therefore, optimizing water pump operations has a significant impact on reducing greenhouse gas emissions and enhances the overall security and reliability of WDNs and PDNs, which motivates the paradigm of operating the two infrastructures in joint fashion [2], [3]. In this context, this paper introduces a framework to explore an optimal strategy for the joint management of water and power distribution networks.

The key decision-making problem involved in the majority of WDN operations, the optimal water flow (OWF) problem, involves solving for optimal pump scheduling given water demand forecasts. The WDN constraints such as pipe and pump hydraulic constraints render the OWF problem a difficult mixed-integer nonlinear program (MINLP). To overcome these challenges, many research efforts have been devoted to modeling, planning, analysis, and operation of WDNs [4]. Similar to the studies on WDNs, there is also a significant amount of work on methods to solve the optimal power flow (OPF) in PDNs, see e.g., [5] for a review. Furthermore, a joint effort between the PDN and the WDN operators facilitates an excellent opportunity to reduce peak energy demand, thereby minimizing both the active power losses and violation of hard

constraints in PDNs (e.g., voltage limits). Studies focusing on the integration of the two networks are discussed next.

The works in [6], [7] explore the flexibility offered by the pumps in WDNs for costs reduction in power transmission networks (rather the PDNs). A power-water flow model incorporating energy flexibility provided by water desalination and treatment plants is proposed in [8], taking into account the constraints of PDN. Refs. [9], [10] develop algorithms to obtain the near-optimal solution to the optimal water-power flow (OWPF) problem. Simultaneous management of thermal and water demands by exploiting the high-inertia thermal dynamics and lossless water storage is proposed in [11]. In [12], the nonconvex chance-constrained water pumping problem has been formulated considering water demand uncertainty subject to both PDN and WDN constraints. The latter work is further expanded in [13] to consider uncertainty in both water and power demand. Ref. [14], formulates an MINLP-based co-optimization model to minimize the operational cost in an islanded micro water-energy nexus, and the MINLP is further transformed into a mixed-integer linear program (MILP) in [15]. A hierarchical framework for the optimization of water and power distribution systems is proposed in [16], where the OWF problem is solved using a genetic algorithm interfaced with EPANET.

Although the integration of power and water distribution systems has been investigated in the previously cited literature, technical limitations still remain. Specifically, most of the previously mentioned works assume linear head loss model for pumps (e.g., [10], [11], and [17]), lack the on/off decisions for pump operation (e.g., [6], [7], [8], and [13]), or assume prior knowledge of water flow directions (e.g., [9], [13]). The present work offers novel technical solutions to the aforementioned limitations.

Our previous work on WDN operations [18]–[20] has contributed linearizations through novel monomial approximations of the WDN hydraulics, but the coupling with the PDN is not explored. Specifically, [18] deals with the water flow problem without pump scheduling, while [20] develops state estimation methods of WDNs. Pump scheduling is investigated in [19] using a heuristic algorithm, without introducing integer variables or co-optimizing the PDN.

The contributions of this paper are as follows: (1) We expand our previous works in [18]–[20] featuring novel monomial approximations for the WDN hydraulics to incorporate pump operations in WDNs using integer variables, cast the OWF problem as MILP, and lift the limitations mentioned in the prior art. It is worth emphasizing that the required number of integer variables is significantly reduced compared to approaches in the prior art. (2) The MILP problem in WDNs is augmented to include the LinDistFlow approximation in PDNs and to develop convex optimal water power flow solver (C-OWPF) for the nonconvex co-optimization of WDNs and PDNs. (3) The C-OWPF solver's performance is assessed in terms of joint objective improvements. (4) The C-OWPF solver's accuracy and feasibility is demonstrated using benchmark nonlinear water and power flow solvers. Besides the technical advantages of the

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C-OWPF solver, this work also investigates the voltage-regulation improvement in PDNs using the co-optimization framework.

The organization of the paper is given next. Section II discusses the modeling of WDNs and PDNs. The OWPF problem is formulated in Section III as a nonlinear and nonconvex program. Section IV presents the proposed approach and casts the nonconvex OWPF problem into a convex OWPF problem, which can be solved as mixed-integer quadratic constrained quadratic programming (MIQCQP) problem. The simulation results are presented in Section V, and the conclusions are drawn in Section VI.

II. WATER AND POWER DISTRIBUTION NETWORK MODELS

The WDN is modeled by a directed graph $(\mathcal{M}, \mathcal{L})$ where $\mathcal{M} = \{0, \dots, M\}$ is a set of $M + 1$ nodes with $\mathcal{M} = \mathcal{J} \cup \mathcal{R} \cup \mathcal{K}$, and $\mathcal{J}$, $\mathcal{R}$, and $\mathcal{K}$ are respectively denote sets of junctions, reservoirs, and tanks. Let $\mathcal{L} \subseteq \mathcal{M} \times \mathcal{M}$ be the set of links connecting the nodes partitioned as $\mathcal{L} = \mathcal{P} \cup \mathcal{W}$, where $\mathcal{P}$ and $\mathcal{W}$ respectively denote the sets of pumps and pipes. Also, in this work, we assume that the pumps are connected to fixed-speed motors, i.e., they are fixed-speed pumps (FSPs) and operate at a constant wire-to-water efficiency η_{ww} . Furthermore, the WDN operation is optimized during an interval t , with $t = \{1, \dots, T\} \in \mathcal{T}$, with δ representing the time interval between two consecutive time periods. Next, we present the mathematical model of WDN operational constraints:

$$\sum_{i:ij \in \mathcal{L}} f_{ij,t} - \sum_{k:j \in \mathcal{L}} f_{jk,t} = d_{j,t}, \quad j \in \mathcal{J} \quad (1a)$$

$$h_{ij,t} = A_{ij} |f_{ij,t}|^{\mu-1} f_{ij,t}, \quad ij \in \mathcal{W} \quad (1b)$$

$$h_{k,t} = h_{k,t-1} + \frac{\delta_t}{A_k} \left(\sum_{i:ik \in \mathcal{L}} f_{ik,t} - \sum_{j:jk \in \mathcal{L}} f_{jk,t} \right), \quad k \in \mathcal{K} \quad (1c)$$

$$h_{ij,t} = \begin{cases} -(h_{0,ij} - \sigma_{ij} f_{ij,t}^{\nu_{ij}}), & \text{if } x_{ij,t} = 1 \\ 0, & \text{if } x_{ij,t} = 0 \end{cases}, \quad ij \in \mathcal{P} \quad (1d)$$

$$f_{ij,\min} x_{ij,t} \leq f_{ij,t} \leq f_{ij,\max} x_{ij,t}, \quad ij \in \mathcal{P} \quad (1e)$$

$$p_{ij,t}^{\text{pump}} = \frac{\rho g}{\eta_{ij,ww}} |f_{ij,t}| f_{ij,t}, \quad ij \in \mathcal{P} \quad (1f)$$

$$h_{i,t} = h_i^R, \quad i \in \mathcal{R} \quad (1g)$$

$$h_{j,\min} \leq h_{j,t} \leq h_{j,\max}, \quad j \in \mathcal{J}, \mathcal{K} \quad (1h)$$

$$\forall t \in \mathcal{T}$$

The mass conservation constraint at junction $j \in \mathcal{J}$ at time t is given in (1a), where $d_{j,t}$ is the estimated water demand at time t . Constraint (1b) formulates the head loss in pipe $ij \in \mathcal{W}$ which is approximated by the empirical Hazen-Williams equation [21], where $A_{ij} = 4.727 C_{ij}^{-1.852} \hat{d}_{ij}^{-4.871} \hat{l}_{ij}$; $\hat{d}_{ij}$ and $\hat{l}_{ij}$ are respectively the diameter and length of a circular pipe; C_{ij} is the Hazen-Williams roughness coefficient (unitless); μ is the flow exponent; and $f_{ij,t}$ is the volumetric flow rate through pipe $ij \in \mathcal{W}$. Constraint (1c) models the water tank head dynamics for tank $k \in \mathcal{K}$ with cross-sectional area A_k and initial head $h_{k,0}$ (assumed to be known). Constraint (1d) formulates the head loss across the FSP connected between nodes i and j and is modeled according to [21, Ch. 3], where $h_{0,ij}$ is the shutoff head for the pump, σ_{ij}, ν_{ij} are the pump curve coefficients evaluated at nominal speed, $f_{ij,t}$ is the flow through the pump and $x_{ij,t}$ is the binary variable to indicate whether the pump $ij \in \mathcal{P}$ is running at time t . The pump flow is constrained by (1e). The FSP power consumption is captured in (1f), where ρ , g , and η_{ij} are respectively the water density, standard gravity coefficient and pump efficiency. Finally, (1g) and (1h) are system constraints. The binary variable $x_{ij,t}$ is set to 1 when pump $(ij) \in \mathcal{P}$ is on at time t and the constraint (1d) is active; otherwise no constraint exists between $h_{i,t}$ and $h_{j,t}$. To accomplish this task

we utilize the big- M technique to rewrite constraint (1d), which is not shown in this paper due to space limitations.

The PDN model is considered next. In particular, the single-feeder radial PDN is modeled using the LinDistFlow approximation of AC power flow equations [22] integrated with PV generation and pumps. Let $\mathcal{N}_0 := \{0, \dots, N\}$ be the set of all buses and $\mathcal{E} := \{1, \dots, N\}$ be the set of lines. The substation bus (root node) is indexed as node 0. The set $\mathcal{N} := \{1, \dots, N\}$ denotes non-substation buses, which represent user buses. Each user bus $n \in \mathcal{N}$ has a unique parent π_n connected via line n . Let $\mathcal{C}_n$ denote set of children nodes corresponding to bus n . For each bus $n \in \mathcal{N}_0$, let $v_{n,t}$ denote its squared voltage magnitude at time t and $(p_{n,t}, q_{n,t})$ denote the real and reactive power injected into bus n . For each line $n \in \mathcal{E}$, let r_n and x_n denote the line series resistance and reactance. Let $p_{n,t}^c$ and $q_{n,t}^c$ denote the uncontrollable active and reactive loads at bus $n \in \mathcal{N}$. The detailed power flow model is given as

$$P_{n,t} = \sum_{j' \in \mathcal{C}_n} P_{j',t} - p_{n,t}, \quad Q_{n,t} = \sum_{j' \in \mathcal{C}_n} Q_{j',t} - q_{n,t} \quad (2a)$$

$$v_{n,t} = v_{\pi_n,t} - 2[r_n P_{n,t} + x_n Q_{n,t}], \quad n \in \mathcal{N}, t \in \mathcal{T}. \quad (2b)$$

$$|q_{i,t}^{\text{pv}}| \leq \sqrt{(s_{i,\max}^{\text{pv}})^2 - (p_{i,t}^{\text{pv}})^2}, \quad i \in \mathcal{N}^{\text{pv}} \subseteq \mathcal{N}, t \in \mathcal{T} \quad (2c)$$

$$p_{n,t} = p_{n,t}^{\text{pv}} - p_{ij,t}^{\text{pump}} - p_{n,t}^c, \quad n \in \mathcal{N}, t \in \mathcal{T} \quad (2d)$$

$$q_{n,t} = q_{n,t}^{\text{pv}} + q_n^{\text{cap}} \left(\frac{v_{n,t}}{v_0} \right) - q_{ij,t}^{\text{pump}} - q_{n,t}^c, \quad n \in \mathcal{N}, t \in \mathcal{T} \quad (2e)$$

$$v_{\min} \leq v_{n,t} \leq v_{\max}, \quad n \in \mathcal{N}, t \in \mathcal{T} \quad (2f)$$

Constraint (2a) describes the active and reactive power flows over line $n \in \mathcal{E}$. The voltage drop or rise in (2b) stems from Ohm's law. The capacity $s_{i,\max}^{\text{pv}}$ of PV inverter $i \in \mathcal{N}^{\text{pv}}$ with active power generation $p_{i,t}^{\text{pv}}$ and reactive power generation/absorption $q_{i,t}^{\text{pv}}$ is included in (2c). The net active and reactive power injections at bus $n \in \mathcal{N}$ are analyzed in (2d) and (2e), where $p_{ij,t}^{\text{pump}}$ and $q_{ij,t}^{\text{pump}}$ are auxiliary optimization variables modeling the active and reactive power consumption of a pump $ij \in \mathcal{P}$ connected to bus n . The reactive power generated by a shunt capacitor at bus n and at nominal voltage is denoted by q_n^{cap} . Constraint (2f) ensures voltage regulation dictated by ANSI C.84.1.

III. CO-OPTIMIZATION OF POWER AND WATER DISTRIBUTION NETWORKS

The pertinent PDN and WDN objective functions are presented next, followed by the OWPF problem formulation.

PDN Objective Function: In this work, we aim to minimize the active power losses across the lines of the PDN. The cost function for the PDN operation is

$$\hat{C}_{n,t}^{\text{loss}}(P_{n,t}, Q_{n,t}) \approx \sum_{n=1}^N \frac{r_n}{v_0} (P_{n,t}^2 + Q_{n,t}^2) \quad (3)$$

It follows from (3) that the active power losses are quadratic in line flows $(P_{n,t}, Q_{n,t})$.

WDN Objective Function: Let $C_{ij,t}^{\text{pump}}$ denote the WDN objective, which is the power required to operate the FSPs [10]. The WDN cost function of the pump $ij \in \mathcal{P}$ at time t is given as

$$C_{ij,t}^{\text{pump}} = p_{ij,t}^{\text{pump}} \quad (4)$$

Note that other pertinent objectives for WDNs can be considered in (4), such as water consumption costs, pump maintenance costs, and water delivery costs; see e.g., [8]. However, the pump power is typically the main contributing factor in WDN costs. Therefore, this work focuses at minimizing pump power.

The optimal water-power flow problem (OWPF) (P1) is a bi-

objective optimization formulated as

$$(P1) \min \sum_{t=1}^T \left[\lambda \hat{C}_{n,t}^{\text{loss}}(P_{n,t}, Q_{n,t}) + (1-\lambda) \sum_{ij \in \mathcal{P}} C_{ij,t}^{\text{pump}} \right] \quad (5a)$$

$$\text{over } \{P_{n,t}, p_{n,t}, Q_{n,t}, q_{n,t}, \{q_{n,t}^{\text{pv}}\}_{\forall n \in \mathcal{N}^{\text{pv}}}, v_{n,t}\}_{t=1}^T \\ \{p_{ij,t}^{\text{pump}}, f_{ij,t}\}_{t=1}^T, \\ \{h_{ij,t}, h_{j,t}\}_{t=1}^T, \quad ij \in \mathcal{L}, j \in \mathcal{M}$$

$$\text{subj. to } (1) - (2) \quad (5b)$$

where λ and $(1-\lambda)$ represent the weight coefficients for the PDN and WDN objectives, respectively. The OWPF problem (P1) is a mixed-integer nonconvex program. The nonconvexities in the continuous variables stem from the head loss models of pipes and pumps (1b) and (1d). Besides, the objective corresponding to pump power consumption (1f) is not convex. Therefore, the pump power (active and reactive) cannot be easily included in the power balance constraint of PDN [cf. (2d) and (2e)] as an equality constraint. To address the above issues, we utilize novel successive linear approximations for head loss models of pipes and pumps in WDNs [20], which solve a sequence of convex problems without prior knowledge of water flow directions in WDNs. These approximations are coupled with the LinDistFlow equations in PDNs, resulting in the successive linear approximation based convex optimal water-power flow (C-OWPF) solver. This is the theme of the following section.

IV. LINEAR APPROXIMATIONS OF NONLINEAR HYDRAULICS IN WDNs AND C-OWPF PROBLEM

First, using a monomial approximation in [19], the nonlinear head loss model for pipes and pumps (1b) and (1d) are approximated around a given point, i.e., flow and can be conveniently represented in linear form. Second, we construct an upper bound for the FSP power consumption (1f) around a given point to represent it in a linear form. Since the resulting approximation of the FSP power consumption is linear, it can be easily included in [cf. (2d) and (2e)] as an equality constraint. These approximations are introduced next.

Linear Approximation of Pipe and Pump Headloss Functions: Using [20] and [19], we replace the nonlinear pipe and pump head loss constraint [c.f (1b) and (1d)] with their corresponding linear form as given below

$$\hat{h}_{ij,t} = \kappa_{ij,t} + f_{ij,t}, \quad ij \in \mathcal{W}, \quad t \in \mathcal{T} \quad (6a)$$

$$\hat{h}_{ij,t} = (\tau_{ij,t} + \varphi_{ij,t} f_{ij,t}), \quad ij \in \mathcal{P}, \quad t \in \mathcal{T} \quad (6b)$$

Constraint (6a) represents the linear head loss approximation for nonlinear pipe head loss constraint (1b), where $\kappa_{ij,t} = \langle f_{ij,t} \rangle (A_{ij} |\langle f_{ij,t} \rangle|^{\mu-1} - 1)$. Similarly, constraint (6b) describes the linear head loss approximation for (1d), with constants $\tau_{ij,t} = -h_{0,ij}$ and $\varphi_{ij,t} = \sigma_{ij} \langle f_{ij,t} \rangle^{\nu_{ij}-1}$. It should be noted that $\kappa_{ij,t}$ and $\varphi_{ij,t}$ are approximated using the flow value $\langle f_{ij,t} \rangle$ through pipes and pumps of the previous iteration. Also, (6b) is valid when the pump is on. Therefore we replace the nonlinear pump head loss constraint (1d) with its corresponding linear form (6b) while implementing the big- M method for the pump head loss constraints.

FSP Pump Power in Linear Form: Attention is then turned to the nonlinear FSP power consumption. Substituting (1d) (when $x_{ij,t} = 1$) in (1f), the FSP power is given as

$$p_{ij,t}^{\text{pump}}(f_{ij,t}) = \eta_{ij} (h_{0,ij} f_{ij,t} - \sigma_{ij} f_{ij,t}^{\nu_{ij}+1}), \quad ij \in \mathcal{P} \quad (7)$$

where $\eta_{ij} = \frac{\rho g}{\eta_{ij, \text{wv}}}$. Since $(p^{\text{pump}}(f_{ij,t}))'' \leq 0$, the FSP power consumption from (7) is concave. Hence, we devise the linear upper

TABLE I: Vector variables to solve C-OWPF problem

Notation	Description
$\mathbf{h}_t$	$(h_{j,t}), j \in \mathcal{M}$
$(\mathbf{h}_t, \mathbf{f}_t)$	$(h_{ij,t}, f_{ij,t}), ij \in \mathcal{L}$
$\mathbf{d}_t$	$(d_{ij,t}), j \in \mathcal{J}$
$(\hat{\mathbf{p}}_t^{\text{pump}}, \mathbf{x}_t)$	$(\hat{p}_{ij,t}^{\text{pump}}, x_{ij,t}), ij \in \mathcal{P}$
κ_t	$(\kappa_{ij,t}), ij \in \mathcal{W}$
$(\alpha_t, \beta_t, \tau_t, \psi_t)$	$(\alpha_{ij,t}, \beta_{ij,t}, \tau_{ij,t}, \psi_{ij,t}), ij \in \mathcal{P}$
$\mathbf{h}_0$	$(h_{k,0}), k \in \mathcal{K}$
$(\mathbf{q}_t^{\text{pv}}, \mathbf{p}_t^{\text{pv}}, \mathbf{s}_{\text{max}}^{\text{pv}})$	$(q_{i,t}^{\text{pv}}, p_{i,t}^{\text{pv}}, s_{i,\text{max}}^{\text{pv}}), i \in \mathcal{N}^{\text{pv}}$
$(\mathbf{P}_t, \mathbf{Q}_t)$	$(P_{n,t}, Q_{n,t}), n \in \mathcal{N}$
$(\mathbf{p}_t, \mathbf{q}_t, \mathbf{v}_t)$	$(p_{n,t}, q_{n,t}, v_{n,t}), n \in \mathcal{N}$
$\mathbf{o}$	Operational limits of WDN and PDN given by (1e), (1g), (1h), and (2f).
$\mathbf{e}_t$	A vector collecting $\mathbf{h}_t, \mathbf{f}_t, \mathbf{x}_t$
$\hat{\mathbf{e}}_t$	A vector collecting $\kappa_t, \tau_t, \psi_t, \alpha_t$, and β_t .

bound to (7) as follows

$$\hat{p}_{ij,t}^{\text{pump}}(f_{ij,t}) = \eta_{ij} (\alpha_{ij,t} f_{ij,t} + \beta_{ij,t} x_{ij,t}) \quad (8)$$

where $\alpha_{ij,t}, \beta_{ij,t}$ are constants which are computed from the previous iteration pump flows $\langle f_{ij,t} \rangle$. Since $\hat{p}_{ij,t}^{\text{pump}}, ij \in \mathcal{P}$ is linear (8), we replace $p_{ij,t}^{\text{pump}}$ and $q_{ij,t}$ in (2d) and (2e) with $\hat{p}_{ij,t}^{\text{pump}}$ and $\hat{q}_{ij,t}^{\text{pump}}$ (assuming a constant power factor).

The convexified optimal water-power flow problem (C-OWPF) for solving the nonconvex OWPF is presented next:

$$(P2) \min \sum_{t=1}^T \left[\lambda \hat{C}_{n,t}^{\text{loss}}(P_{n,t}, Q_{n,t}) + (1-\lambda) \sum_{ij \in \mathcal{P}} \hat{C}_{ij,t}^{\text{pump}} \right] \quad (9a)$$

$$\text{over } \{P_{n,t}, p_{n,t}, Q_{n,t}, q_{n,t}, \{q_{n,t}^{\text{pv}}\}_{\forall n \in \mathcal{N}^{\text{pv}}}, v_{n,t}\}_{t=1}^T \\ \{\hat{p}_{ij,t}^{\text{pump}}, f_{ij,t}\}_{t=1}^T, \\ \{\hat{h}_{ij,t}, h_{j,t}\}_{t=1}^T, \quad ij \in \mathcal{L}, j \in \mathcal{M} \\ \text{subj. to } (1a), (1c), (1e), (1g), (1h), (2), (6), (8) \quad (9b)$$

where $\hat{C}_{ij,t}^{\text{pump}}$ is given by (8). Notice that the C-OWPF problem (P2) can be solved as mixed-integer quadratically constrained quadratic program (MIQCQP) problem. We use the term convex in C-OWPF to emphasize that *the continuous part of (9) is convex*. It is worth noting that modern mixed integer convex programming software can reliably solve instances of (9).

Remark. *Computational complexity of C-OWPF:* The original OWPF problem (P1) is an MINLP with $|\mathcal{J}| + |\mathcal{L}| + 3|\mathcal{N}| + |\mathcal{N}^{\text{pv}}| + 2|\mathcal{E}|$ continuous variables and $|\mathcal{P}|$ binary variables and can be solved with significant computational burden [7], [15], [23]. On the other hand, the C-OWPF problem (P2) is an MIQCQP that includes the same amount of continuous and binary variables but can be solved with significant computational savings, because the continuous part is convex. Furthermore, the works in [7], [8], [15] include an MINLP of expanded size with $|\mathcal{P}| + |\mathcal{W}|$ binary variables (where $|\mathcal{W}| \gg |\mathcal{P}|$ in typical WDNs). The reason is that binary variables are used to account for water flow directions in pipes. The C-OWPF needs only $|\mathcal{P}|$ binary variables because it yields the direction of water flow in pipes without the need for additional binary variables.

C-OWPF Iterative Algorithm: The smart city operator implements the solution to the C-OWPF problem. The detailed implementation is summarized in Algorithm 1. For notational convenience, we define the relevant vectors for WDNs and PDNs in Table I.

Comparisons with Decoupled Designs: To evaluate the efficiency of the proposed C-OWPF, we compare it with a decoupled design. In particular, the decoupled design consists of two stages. In

Algorithm 1: C-OWPF Iterative Algorithm

Input : Smart city operator acquires WDN and PDN topologies, operational limits $\mathbf{o}$, tanks initial condition $\mathbf{h}_0$, $\{\mathbf{d}_t, \text{ and forecasts } \mathbf{p}_t^c, \mathbf{q}_t^c, \mathbf{p}_t^{pv}\}_{t=1}^T$

Output: $\langle \{\varepsilon_t\}_{t=1}^T \rangle_{\text{final}}$

- 1 Initialize $l = 0$ and $\langle \{\varepsilon_t\}_{t=1}^T \rangle_0$;
- 2 Set $\langle \{\varepsilon_t\}_{t=1}^T \rangle_{\text{save}} = \langle \{\varepsilon_t\}_{t=1}^T \rangle_0$, $l = 1$, maxIter ;
- 3 **while** error $\geq$ tolerance **OR** $l \leq \text{maxIter}$ **do**
- 4 Obtain $\langle \{\hat{\varepsilon}_t\}_{t=1}^T \rangle_l$ from $\langle \{\varepsilon_t\}_{t=1}^T \rangle_{\text{save}}$;
- 5 Solve problem (P2) over $\{\mathbf{P}_t, \mathbf{Q}_t, \mathbf{p}_t, \mathbf{q}_t, \mathbf{q}_t^{pv}, \mathbf{v}_t\}_{t=1}^T$, $\{\hat{\mathbf{p}}_t^{\text{pump}}, \mathbf{f}_t\}_{t=1}^T$, $\{\mathbf{h}_t, \hat{\mathbf{h}}_t, \mathbf{x}_t\}_{t=1}^T$; extract $\langle \{\varepsilon_t\}_{t=1}^T \rangle_l$;
- 6 Calculate error := $\text{norm}(\langle \{\varepsilon_t\}_{t=1}^T \rangle_l - \langle \{\varepsilon_t\}_{t=1}^T \rangle_{\text{save}})$;
- 7 Update $\langle \{\varepsilon_t\}_{t=1}^T \rangle_{\text{save}} = \langle \{\varepsilon_t\}_{t=1}^T \rangle_l$, and $l = l + 1$
- 8 **end**
- 9 Set $\langle \{\varepsilon_t\}_{t=1}^T \rangle_l = \langle \{\varepsilon_t\}_{t=1}^T \rangle_{\text{final}}$;
- 10 Smart city operator communicate setpoints $\{\mathbf{q}_t^{pv}\}_{t=1}^T$ to PDN operators, and $\{\mathbf{x}_t\}_{t=1}^T$ to WDN operators.

stage I, the WDN operator first solves OWF problem and then inputs the optimal pump powers (active & reactive power) to the PDN operator. This design is similar to that discussed in [7]. To solve the OWF problem in stage I, the WDN operator minimizes $\sum_{t=1}^T \sum_{ij \in \mathcal{P}} \hat{C}_{ij,t}^{\text{pump}}$ subjected to linear WDN constraints, wherein the variables related to WDN are updated according to the iterative algorithm in [18]. Therefore, the OWF problem in stage I is a mixed-integer linear programming (MILP) problem. In stage II, the PDN operator solves OPF problem by minimizing $\sum_{t=1}^T \hat{C}_{n,t}^{\text{loss}}(\mathbf{P}_{n,t}, \mathbf{Q}_{n,t})$ subjected to PDN constraints, which is a quadratically constrained quadratic program (QCQP). This problem yields the optimal PV inverter reactive power setpoints. For easier reference, we refer to the decoupled problem, that is, the sequence of stage I and stage II problems, as Scenario A; and the C-OWPF problem (P2) is referred as Scenario B. It is worth noting that, although we use linear approximations in Scenarios A and B, we utilize the exact nonlinear solvers EPANET for WDNs [21] and the Z-bus method [24] for PDNs to evaluate the feasibility and modeling accuracy of the C-OWPF solver.

V. NUMERICAL RESULTS

This section examines the C-OWPF solver's performance in terms of joint objective improvements over the decoupled design.

The case study considered in this paper consists of the modified IEEE-13 bus PDN [25] and 8-node WDN [21] depicted in Fig. 1. We operate the system over a time horizon of $T = 12$ hours, beginning at 8 AM with δ as one hour. The optimizations are solved using the MATLAB-based optimization toolbox CVX along with the mixed-integer solver Gurobi [26], [27]. All simulations are run on a 3.60-GHz, intel core i7 computer with 32 GB of RAM. For WDNs, the head unit is feet [ft]; and the flow unit is gallons per minute [GPM]. For PDNs, V_{base} is 2.410 kV and S_{base} is 5 MVA. Fig. 2 illustrates the network-wide water and power demands, which are carefully chosen to reflect reasonable conditions.

For the 8-node WDN, the pipe lengths, diameters, node elevations, tank diameter, and junction base demands are obtained from [21]. The bounds on pump flows chosen to be $[f_{\min}, f_{\max}] = [0, 2400]$ GPM. The tank minimum height, maximum height, and initial level are respectively set to $h_{\min} = 830$ ft, $h_{\max} = 850$ ft, and $h_0 = 840$ ft. Furthermore, we consider an FSP with parameters $h_0 = 266.67$ ft, $\sigma = 4.629 \times 10^{-5}$, and $\nu = 2$ computed from EPANET [21]. Similarly, for the modified 13-bus PDN, the line

parameters and nominal loads obtained from [28], wherein five PV inverters are assumed to be located at buses {634, 645, 646, 692, 652}. Furthermore, we assume s_{max}^{pv} to be 170% of p_{max}^{pv} . The bounds on the squared magnitude voltage (2f) are respectively set to be $(0.956)^2$ and $(1.05)^2$ (p.u.)².

Next, we define the objective functions as follows: $\sum_t \sum_{ij} (C_{ij,t}^{\text{pump}})$ is the *exact* total pump power consumption (kW) computed from (1f); $\sum_t (C_{\text{loss}})$ represent the *exact* total losses (kW) across lines in the PDN which are computed after the power flows are obtained from the Z-bus method [24]. The *Total Objective* (kW) amounts to the sum of active power losses and pump power consumption. Also, the weight coefficient λ is set to 1.0 in Scenario B, i.e., priority is given to the active power loss minimization in PDNs. Even with this simple choice, gains to both networks can be demonstrated. The percent reduction in objective value for the scenarios considered here is computed as follows:

$$\% \text{ reduction} = \frac{\text{Obj. value in } Y_1 - \text{Obj. value in } Y_2}{\text{Obj. value in } Y_1} \quad (10)$$

where Y_1 and Y_2 correspond to Scenarios A and B, respectively.

The resulting scheduling decisions for the two scenarios are presented next. In Scenario A, the pump is turned on during the times when the load on the PDN was high, namely, from 8–9 hours, and also 12–13 hours. The compounding of the pumping and remaining uncontrollable load in this case causes significant voltage drop and renders Scenario A infeasible in stage II. On the other hand, in Scenario B, the pump is turned on when the load on the PDN is small and at the times during high PV generation, i.e., from 14–16 hours, enabling Scenario B to be feasible. This result emphasizes that the coordinated operation of the two systems, i.e., Scenario B, leads to better voltage regulation in PDNs than the Scenario A.

In short, the decoupled operation violates the lower voltage regulation bound in (2f). Due to this infeasibility in scenario A, and in order to be able to provide a comparison of operational costs, the PV inverter setpoints are computed upon dropping the voltage regulation constraint (2f) in scenario A.

The % reductions in objective values are listed in Table II. It can be seen that the C-OWPF solver (Scenario B) achieved an actual reduction in all the objective values when compared to the decoupled approach (Scenario A). For example, the *total objective* value is respectively reduced by 1.00 %. The % reductions in operational costs are primarily achieved due to operating FSPs during times of low water and power demand.

The context around the reported reductions is important here. First, the reduction values in Table II are based on a 12-hour operation, and the savings should eventually be assessed over a sufficiently long period (e.g., annually). More importantly, this comparison has been performed under violation of the voltage regulation constraint in the decoupled operation, but not in the coupled operation. Accounting for the resulting compromise in the network security for the decoupled operation implies significantly higher savings.

The computational time for Scenarios A and B is less than 2 minutes. Finally, the modeling accuracy of the developed linearizations, that is, the flows, heads, and pump power consumption computed using C-OWPF solver, is compared with the benchmark software EPANET, which uses the nonlinear hydraulics. For the present case study, the nodal heads and flows found by the two models differ at most by 0.001 ft and 0.02 GPM, respectively. The pump power computed by the two models differs by 0.05 kW as given in Table III, which also shows that head gain of the pump is also accurately computed. The aforementioned results show that

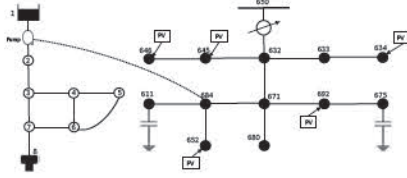


Fig. 1: Water (left) and power (right) distribution networks for the case study. Dashed line shows the pump connection in PDN.

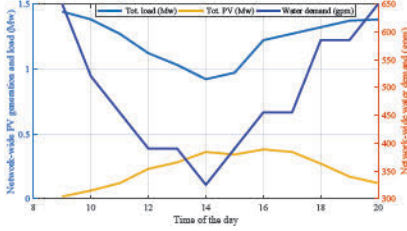


Fig. 2: Network-wide total PV, load, and water demand profiles.

the successive linearization method returns WDN variables that accurately satisfy the nonlinear hydraulics.

VI. CONCLUSIONS

This work develops a novel solver (C-OWPF) to integrate water and power distribution networks. The merits of the proposed approach are demonstrated in terms of joint operational cost and security (voltage regulation) improvements for the pilot study of this work. Future work will look at incorporating the variable-speed pumps (VSPs) in WDNs and the formulation of the stochastic framework to account for the uncertainties of power consumption, water demands, and PV renewable power. Future work will also experiment with larger WDNs and incorporate price information for WDNs and PDNs.

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TABLE II: Comparison of objective function values (in kW) for Scenario A and Scenario B

Objective function	Scenario A	Scenario B	% Reduction
$\sum_t \sum_{ij \in \mathcal{P}} C_{ij,t}^{\text{pump}}$	113.26	111.81	1.28
$\sum_t C_{n,t}^{\text{loss}}$	314.05	311.24	0.89
Total Objective	427.31	423.05	1.00

TABLE III: Pump head gain and power in Scenario B for $x_{ij,t} = 1$

	$h_{ij,t}$ (ft)		$\sum_{ij} P_{ij,t}^{\text{pump}}$ (kW)	
	C-OWPF	EPANET	C-OWPF	EPANET
$t = 14 - 15$ (hrs)	204.891	204.892	55.71	55.76
$t = 15 - 16$ (hrs)	202.636	204.637	56.10	56.15

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